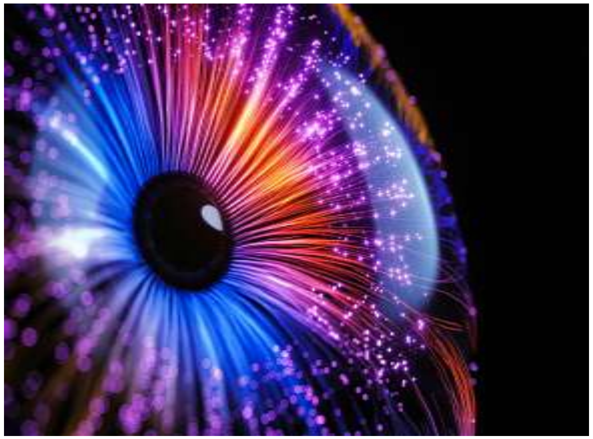
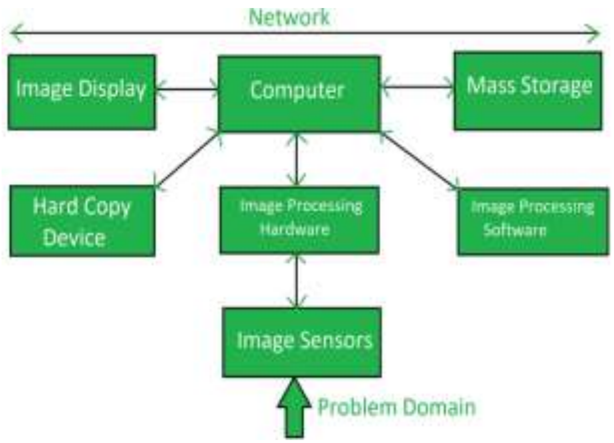
Digital Image Processing





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Topic 7

Image Compression

Introduction

Image compression reduces file size by removing redundant data, enabling efficient storage and transmission, using techniques like lossless (perfect reconstruction) or lossy (slight quality loss for greater reduction) methods, achieved through encoding processes that exploit pixel correlation, often involving transforms like DCT or wavelets to concentrate image energy, followed by quantization and symbol coding, with popular formats including JPEG and PNG.

Purpose

- **Redundancy Reduction:** Exploits patterns in data (e.g., adjacent pixels having similar colors) to represent information more compactly.
- **Efficiency:** Smaller files mean faster loading on websites, less storage space, and quicker internet transfers.

Key Redundancies

Compression algorithms target three main types of redundancy:

- **Coding Redundancy:** Occurs when the codes used for pixel values are not optimal. Using variable-length coding (shorter codes for more frequent values) reduces this.

- **Interpixel (Spatial) Redundancy:** Results from the high correlation between adjacent pixels (e.g., large areas of the same color). Predictive or transform methods are used to reduce this.

- **Psychovisual Redundancy:** Refers to information that the human eye cannot easily perceive. This is "irrelevant" data that can be discarded without significantly impacting the viewer's experience.

Components

Mapper (Transformation): Converts image data (e.g., pixel values) into a new domain (like frequency coefficients using DCT or DWT) where data is more compact.

Quantizer: Reduces the precision of the transformed data, removing less important details (irreversible in lossy compression).

Symbol Encoder: Assigns short codes to frequent patterns for further size reduction (e.g., Huffman coding).

Decoder: Performs inverse operations (Inverse Symbol Decoder, Inverse Mapper) to reconstruct the image, but without the lost precision from quantization in lossy methods.

Types of Compression

Compression techniques are broadly classified into two categories:

- **Lossless Compression:** Allows for the perfect reconstruction of the original image with zero data loss. It is essential for medical imaging, archival, and technical drawings.
  - *Common Algorithms:* Run-Length Encoding (RLE), Huffman Coding, Lempel-Ziv-Welch (LZW).

- **Lossy Compression:** Achieves much higher compression ratios by permanently discarding less significant information. It is ideal for photographs and web images where minor quality loss is acceptable.
  - *Common Algorithms:* Discrete Cosine Transform (DCT), Wavelet Transform, Fractal Compression.

Common Formats

Format	Compression Type	Primary Use
JPEG	Lossy (DCT-based)	Photographs, general web images
PNG	Lossless (Deflate)	Graphics, logos, and images requiring transparency
GIF	Lossless (LZW)	Simple animations and low-color graphics
WebP	Both (Lossy/Lossless)	Modern web optimization (developed by Google)
HEIF/HEIC	Lossy	High-efficiency mobile photography (Apple/standard)

Relative Data Redundancy

- Let b and b' denote the number of bits in two representations of the same information, the relative data redundancy R is

$$R = 1 - 1/C$$

C is called the compression ratio, defined as

$$C = b/b'$$

e.g., C = 10, the corresponding relative data redundancy of the larger representation is 0.9, indicating that 90% of its data is redundant

Necessity of Compression

- Data storage
- Data transmission

The digital information revolution has brought with it an explosive increase in the amount of data.

These digital data requires large storage and high transmission bandwidth. For example:

- 24 hour record of Electrocardiogram (1-D signal) with sampling rate of 360 Hz & 11 bits/sample data resolution requires about 43 Mbytes per channel.
- Finger prints (2-D signal) of a pair of hands with resolution 500 pixels per inch require 6 Mbytes of storage.

Compression is essential to facilitate storage capacity and economical rapid transmission over low bandwidth communication line.

How can we implement compression?

- **Coding redundancy**

  Most 2-D intensity arrays contain more bits than are needed to represent the intensities

- **Spatial and temporal redundancy**

  Pixels of most 2-D intensity arrays are correlated spatially and video sequences are temporally correlated

- **Irrelevant information**

  Most 2-D intensity arrays contain information that is ignored by the human visual system

Examples of Redundancy

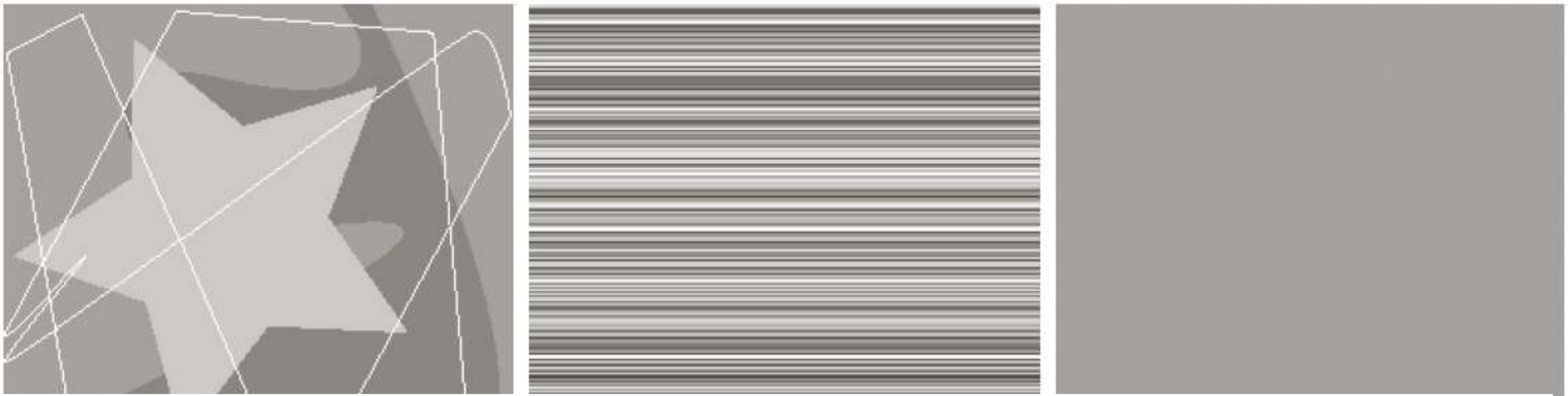


FIGURE 8.1 Computer generated 256 × 256 × 8 bit images with (a) coding redundancy, (b) spatial redundancy, and (c) irrelevant information. (Each was designed to demonstrate one principal redundancy but may exhibit others as well.)

Coding Redundancy

TABLE 8.1
Example of variable-length coding.

r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_{87} = 87$	0.25	01010111	8	01	2
$r_{128} = 128$	0.47	10000000	8	1	1
$r_{186} = 186$	0.25	11000100	8	000	3
$r_{255} = 255$	0.03	11111111	8	001	3
r_k for $k \neq 87, 128, 186, 255$	0	—	8	—	0

The average number of bits required to represent each pixel is

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k)\, p_r(r_k) = 0.25(2) + 0.47(1) + 0.24(3) + 0.03(3) = 1.81 bits$$

$$C = \frac{8}{1.81} \approx 4.42$$

$$R = 1 - 1/4.42 = 0.774$$

Spatial and Temporal Redundancy

1. All 256 intensities are equally probable.
2. The pixels along each line are identical.
3. The intensity of each line was selected randomly.

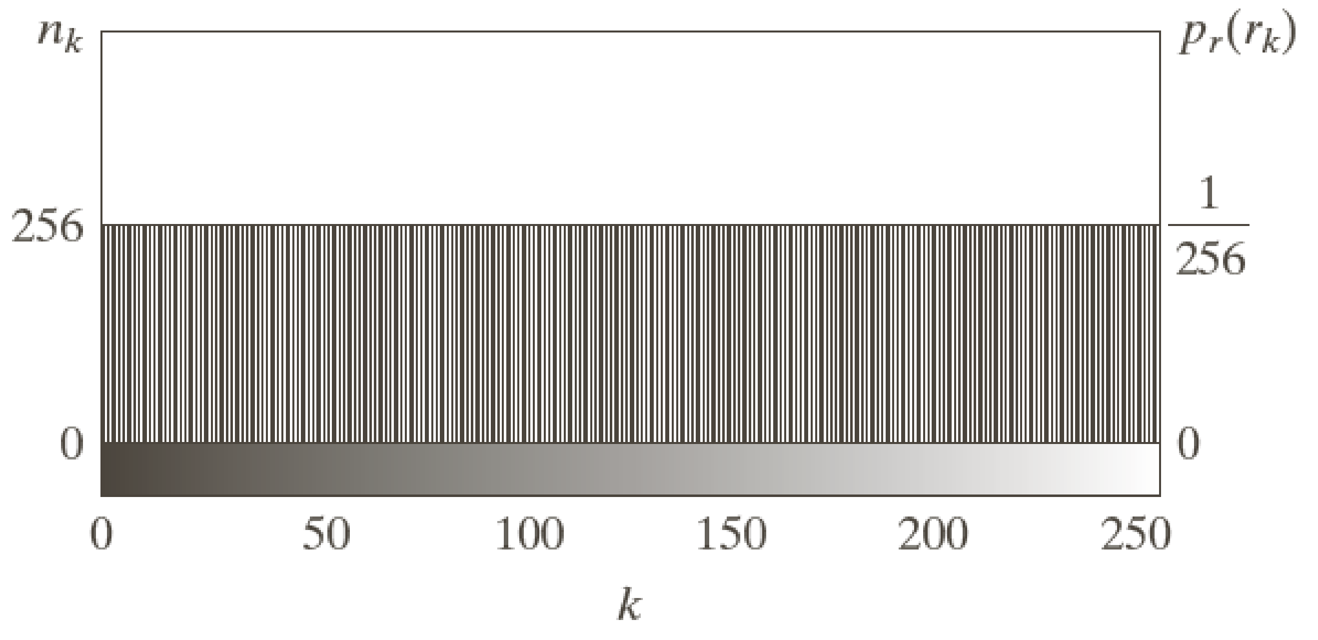


FIGURE 8.2 The intensity histogram of the image in Fig. 8.1(b).

Spatial and Temporal Redundancy

1. All 256 intensities are equally probable.
2. The pixels along each line are identical.
3. The intensity of each line was selected randomly.

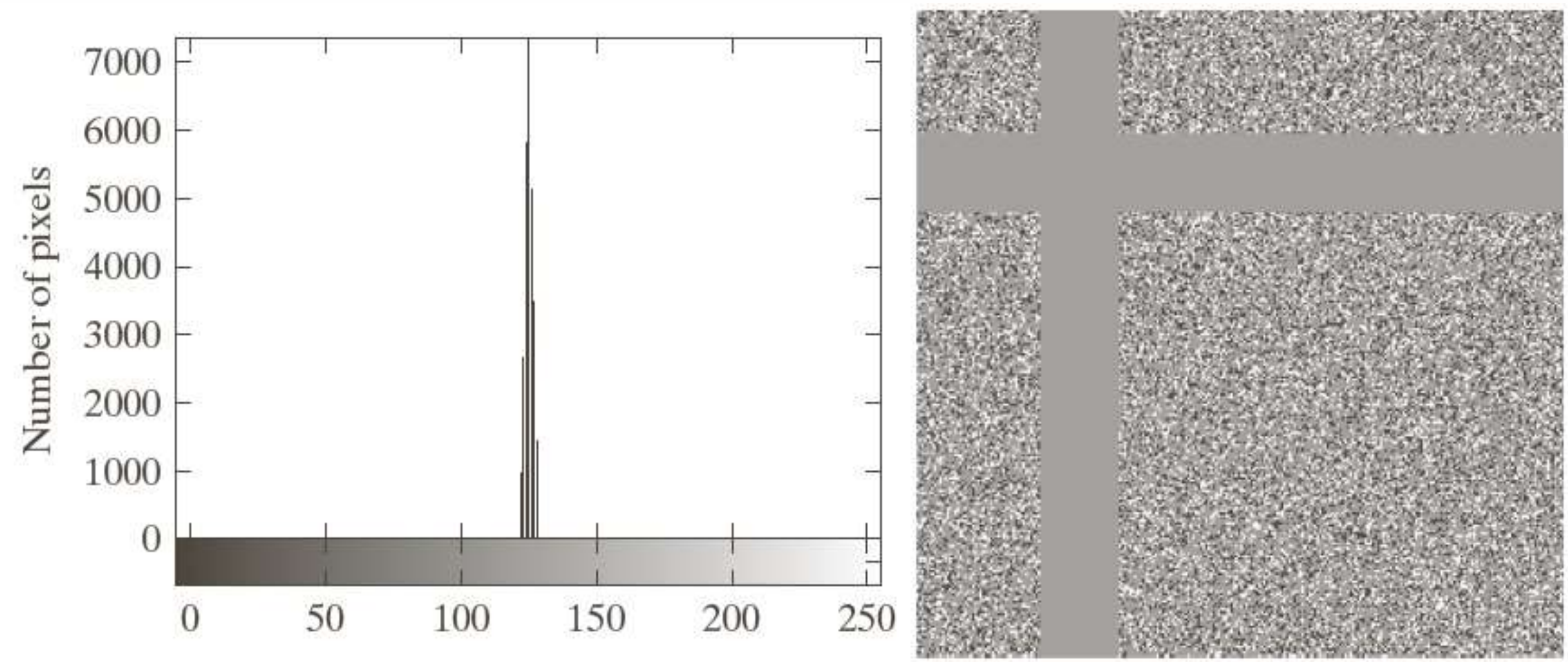
Run-length pair specifies the start of a new intensity and the number of consecutive pixels that have that intensity.

Each 256-pixel line of the original representation is replaced by a single 8-bit intensity value and length 256 in the run-length representation.

The compression ratio is

$$\frac{256 \times 256 \times 8}{(256 + 256) \times 8} = 128:1$$

Irrelevant Information



a b

FIGURE 8.3 (a) Histogram of the image in Fig. 8.1(c) and (b) a histogram equalized version of the image.

$$256 \times 256 \times 8 / 8$$

$$= 65536 : 1$$

Measuring Image Information

A random event E with probability P(E) is said to contain

$$I(E) = \log \frac{1}{P(E)} = -\log P(E)$$

units of information.

Measuring Image Information

Given a source of statistically independent random events from a discrete set of possible events $\{a_1, a_2, \ ..., \ a_J\}$ with associated probabilities $\{P(a_1), \ P(a_2), \ ..., \ P(a_J)\}$, the average information per source output, called the entropy of the source

$$H = -\sum_{j=1}^{J} P(a_j) \log P(a_j)$$

a_j is called source symbols. Because they are statistically independent, the source called $zero-memory \ source.$

Measuring Image Information

If an image is considered to be the output of an imaginary zero-memory "intensity source", we can use the histogram of the observed image to estimate the symbol probabilities of the source. The intensity source's entropy becomes

$$\tilde{H} = -\sum_{k=0}^{L-1} p_r(r_k) \log p_r(r_k)$$

$p_r(r_k)$ is the normalized histogram.

Measuring Image Information

For the fig.8.1(a),

$$\tilde{H} =$$

$$-[0.25\log_2 0.25 + 0.47\log_2 0.47 + 0.25\log_2 0.25 + 0.03\log_2 0.03]$$

$$\approx 1.6614 \text{ bits/pixel}$$

Fidelity Criteria

Let $f(x,y)$ be an input image and $\hat{f}(x,y)$ be an approximation of $f(x,y)$. The images are of size $M \times N$.

The *root-mean-square error* is

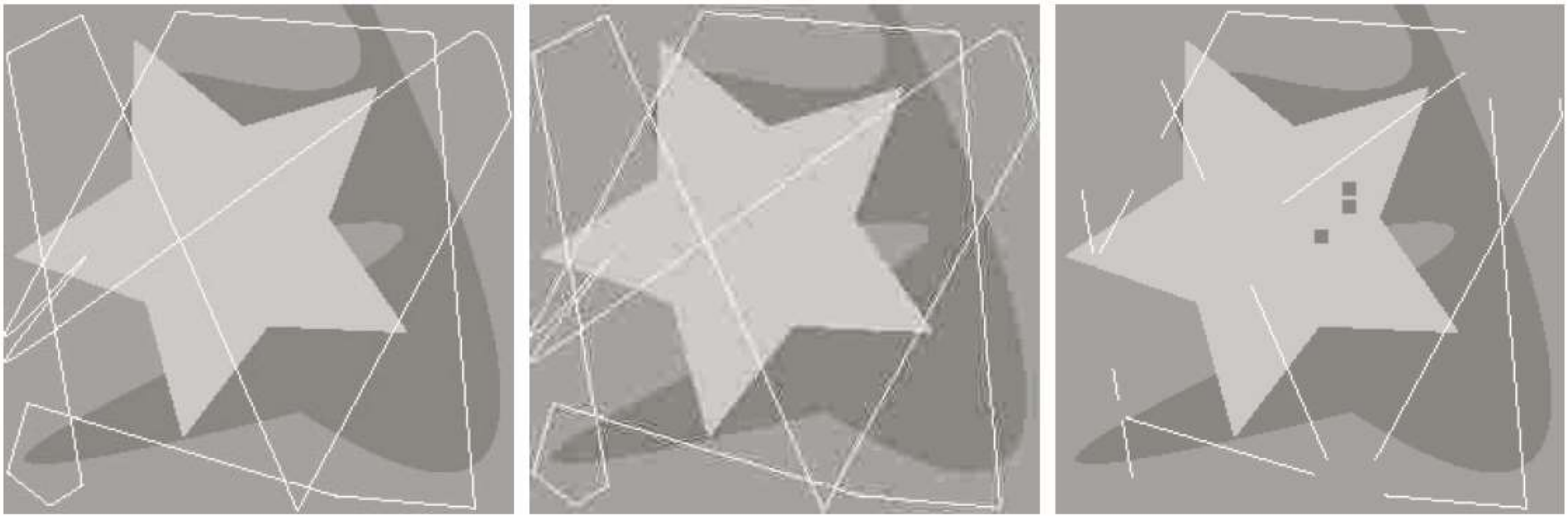
$$e_{rms} = \left[\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} \left[\hat{f}(x,y) - f(x,y) \right]^2 \right]^{1/2}$$

Fidelity Criteria

The *mean-square signal-to-noise ratio* of the output image, denoted $\mathrm{SNR}_{\mathrm{ms}}$

$$\mathrm{SNR}_{\mathrm{ms}} = \frac{\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}\left[\hat{f}(x,y)\right]^2}{\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}\left[\hat{f}(x,y) - f(x,y)\right]^2}$$

RMSE = 5.17 RMSE = 15.67 RMSE = 14.17



a b c

FIGURE 8.4 Three approximations of the image in Fig. 8.1(a).

Image Compression Model

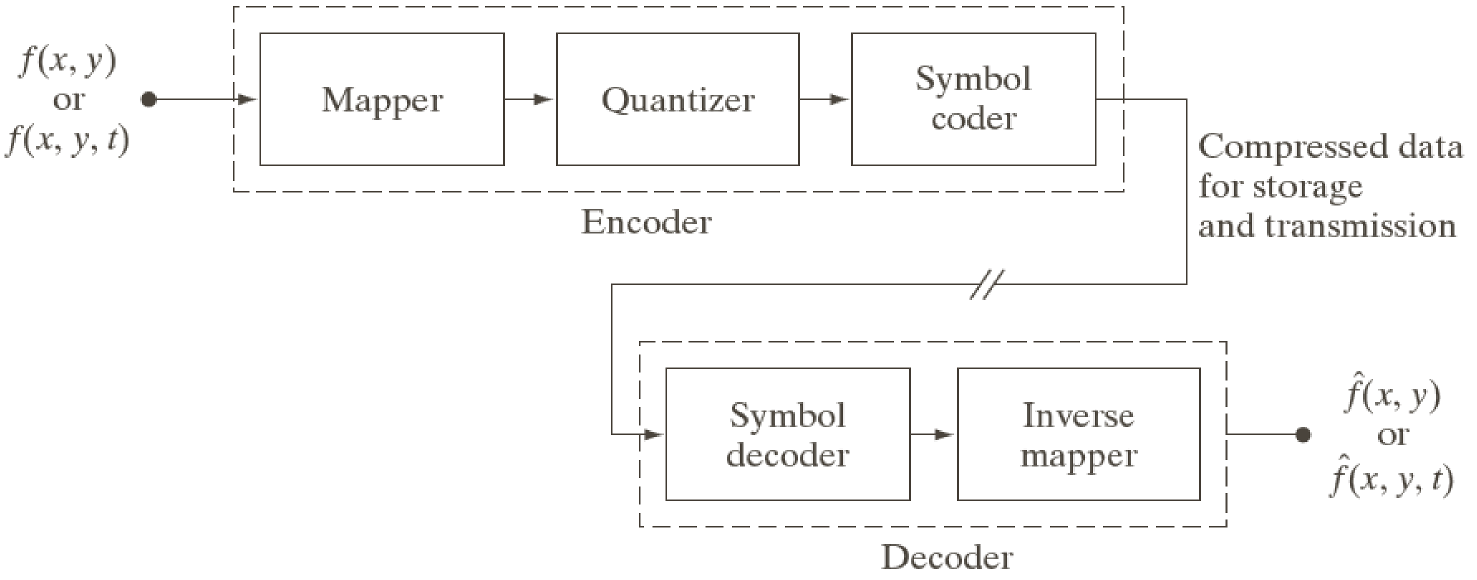


FIGURE 8.5 Functional block diagram of a general image compression system.

Image Compression Standards

Image Compression Standards, Formats, and Containers

- **Still Image**
  - **Binary**
    - CCITT Group 3
    - CCITT Group 4
    - JBIG (or JBIG1)
    - JBIG2
    - TIFF
  - **Continuous Tone**
    - JPEG
    - JPEG-LS
    - JPEG-2000
    - BMP
    - GIF
    - PDF
    - PNG
    - TIFF
- **Video**
  - DV
  - H.261
  - H.262
  - H.263
  - H.264
  - MPEG-1
  - MPEG-2
  - MPEG-4
  - MPEG-4 AVC
  - AVS
  - HDV
  - M-JPEG
  - QuickTime
  - VC-1 (or WMV9)

FIGURE 8.6 Some popular image compression standards, file formats, and containers. Internationally sanctioned entries are shown in black; all others are grayed.

Name	Organization	Description
Bi-Level Still Images		
CCITT Group 3	ITU-T	Designed as a facsimile (FAX) method for transmitting binary documents over telephone lines. Supports 1-D and 2-D run-length [8.2.5] and Huffman [8.2.1] coding.
CCITT Group 4	ITU-T	A simplified and streamlined version of the CCITT Group 3 standard supporting 2-D run-length coding only.
JBIG *or* JBIG1	ISO/IEC/ ITU-T	A *Joint Bi-level Image Experts Group* standard for progressive, lossless compression of bi-level images. Continuous-tone images of up to 6 bits/pixel can be coded on a bit-plane basis [8.2.7]. Context sensitive arithmetic coding [8.2.3] is used and an initial low resolution version of the image can be gradually enhanced with additional compressed data.
JBIG2	ISO/IEC/ ITU-T	A follow-on to JBIG1 for bi-level images in desktop, Internet, and FAX applications. The compression method used is content based, with dictionary based methods [8.2.6] for text and halftone regions, and Huffman [8.2.1] or arithmetic coding [8.2.3] for other image content. It can be lossy or lossless.
Continuous-Tone Still Images		
JPEG	ISO/IEC/ ITU-T	A *Joint Photographic Experts Group* standard for images of photographic quality. Its lossy *baseline coding system* (most commonly implemented) uses quantized discrete cosine transforms (DCT) on 8×8 image blocks [8.2.8], Huffman [8.2.1], and run-length [8.2.5] coding. It is one of the most popular methods for compressing images on the Internet.
JPEG-LS	ISO/IEC/ ITU-T	A lossless to near-lossless standard for continuous tone images based on adaptive prediction [8.2.9], context modeling [8.2.3], and Golomb coding [8.2.2].
JPEG-2000	ISO/IEC/ ITU-T	A follow-on to JPEG for increased compression of photographic quality images. Arithmetic coding [8.2.3] and quantized discrete wavelet transforms (DWT) [8.2.10] are used. The compression can be lossy or lossless.

TABLE 8.3 Internationally sanctioned image compression standards. The numbers in brackets refer to sections in this chapter.

(*Continues*)

TABLE 8.3 (*Continued*)

Name	Organization	Description
Video		
DV	IEC	*Digital Video*. A video standard tailored to home and semiprofessional video production applications and equipment—like electronic news gathering and camcorders. Frames are compressed independently for uncomplicated editing using a DCT-based approach [8.2.8] similar to JPEG.
H.261	ITU-T	A two-way videoconferencing standard for ISDN (*integrated services digital network*) lines. It supports non-interlaced 352 × 288 and 176 × 144 resolution images, called CIF (*Common Intermediate Format*) and QCIF (*Quarter CIF*), respectively. A DCT-based compression approach [8.2.8] similar to JPEG is used, with frame-to-frame prediction differencing [8.2.9] to reduce temporal redundancy. A block-based technique is used to compensate for motion between frames.
H.262	ITU-T	See MPEG-2 below.
H.263	ITU-T	An enhanced version of H.261 designed for ordinary telephone modems (i.e., 28.8 Kb/s) with additional resolutions: SQCIF (*Sub-Quarter* CIF 128 × 96), 4CIF (704 × 576), and 16CIF (1408 × 512).
H.264	ITU-T	An extension of H.261–H.263 for videoconferencing, Internet streaming, and television broadcasting. It supports prediction differences within frames [8.2.9], variable block size integer transforms (rather than the DCT), and context adaptive arithmetic coding [8.2.3].
MPEG-1	ISO/IEC	A *Motion Pictures Expert Group* standard for CD-ROM applications with non-interlaced video at up to 1.5 Mb/s. It is similar to H.261 but frame predictions can be based on the previous frame, next frame, or an interpolation of both. It is supported by almost all computers and DVD players.
MPEG-2	ISO/IEC	An extension of MPEG-1 designed for DVDs with transfer rates to 15 Mb/s. Supports interlaced video and HDTV. It is the most successful video standard to date.
MPEG-4	ISO/IEC	An extension of MPEG-2 that supports variable block sizes and prediction differencing [8.2.9] within frames.
MPEG-4 AVC	ISO/IEC	MPEG-4 Part 10 *Advanced Video Coding* (AVC). Identical to H.264 above.

Name	Organization	Description
Continuous-Tone Still Images		
BMP	Microsoft	*Windows Bitmap*. A file format used mainly for simple uncompressed images.
GIF	CompuServe	*Graphic Interchange Format*. A file format that uses lossless LZW coding [8.2.4] for 1- through 8-bit images. It is frequently used to make small animations and short low resolution films for the World Wide Web.
PDF	Adobe Systems	*Portable Document Format*. A format for representing 2-D documents in a device and resolution independent way. It can function as a container for JPEG, JPEG 2000, CCITT, and other compressed images. Some PDF versions have become ISO standards.
PNG	*World Wide Web Consortium* (W3C)	*Portable Network Graphics*. A file format that losslessly compresses full color images with transparency (up to 48 bits/pixel) by coding the difference between each pixel's value and a predicted value based on past pixels [8.2.9].
TIFF	Aldus	*Tagged Image File Format*. A flexible file format supporting a variety of image compression standards, including JPEG, JPEG-LS, JPEG-2000, JBIG2, and others.
Video		
AVS	MII	*Audio-Video Standard*. Similar to H.264 but uses exponential Golomb coding [8.2.2]. Developed in China.
HDV	Company consortium	*High Definition Video*. An extension of DV for HD television that uses MPEG-2 like compression, including temporal redundancy removal by prediction differencing [8.2.9].
M-JPEG	Various companies	*Motion JPEG*. A compression format in which each frame is compressed independently using JPEG.
Quick-Time	Apple Computer	A media container supporting DV, H.261, H.262, H.264, MPEG-1, MPEG-2, MPEG-4, and other video compression formats.
VC-1 WMV9	SMPTE Microsoft	The most used video format on the Internet. Adopted for HD and *Blu-ray* high-definition DVDs. It is similar to H.264/AVC, using an integer DCT with varying block sizes [8.2.8 and 8.2.9] and context dependent variable-length code tables [8.2.1]—but no predictions within frames.

TABLE 8.4
Popular image compression standards, file formats, and containers, not included in Table 8.3.

Some Basic Compression Methods

1. Huffman Coding

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	0.4
a_1	0.1	0.1	0.2	0.3	
a_4	0.1	0.1	0.1		
a_3	0.06	0.1			
a_5	0.04				

FIGURE 8.7 Huffman source reductions.

Huffman Coding

Original source			Source reduction							
Symbol	Probability	Code	1		2		3		4	
a_2	0.4	1	0.4	1	0.4	1	0.4	1	0.6	0
a_6	0.3	00	0.3	00	0.3	00	0.3	00	0.4	1
a_1	0.1	011	0.1	011	0.2	010	0.3	01		
a_4	0.1	0100	0.1	0100	0.1	011				
a_3	0.06	01010	0.1	0101						
a_5	0.04	01011								

FIGURE 8.8 Huffman code assignment procedure.

The average length of this code is

$$L_{avg} = 0.4*1+0.3*2+0.1*3+0.1*4+0.06*5+0.04*5$$

$= 2.2$ bits/pixel

2. LZW (Dictionary) Coding

LZW (Lempel-Ziv-Welch) coding, assigns fixed-length code words to variable length sequences of source symbols, but requires no *a priori* knowledge of the probability of the source symbols.

LZW is used in:

- *Tagged Image file format (TIFF)*
- *Graphic interchange format (GIF)*
- *Portable document format (PDF)*

LZW was formulated in 1984

LZW Coding: Algorithm

- A codebook or “dictionary” containing the source symbols is constructed.
- For 8-bit monochrome images, the first 256 words of the dictionary are assigned to the gray levels 0-255
- Remaining part of the dictionary is filled with sequences of the gray levels

Important Features of LZW

1. The dictionary is created while the data are being encoded. So encoding can be done "on the fly"
2. The dictionary is not required to be transmitted. The dictionary will be built up in the decoding
3. If the dictionary "overflows" then we have to reinitialize the dictionary and add a bit to each one of the code words.
4. Choosing a large dictionary size avoids overflow, but spoils compressions

3. Run-Length Coding

1. Run-length Encoding, or **RLE** is a technique used to **reduce the size of a repeating string of characters**.
2. This repeating string is called a *run*, typically RLE encodes a run of symbols into two bytes , a **count** and a **symbol**.
3. RLE can compress any type of data.
4. RLE cannot achieve high compression ratios compared to other compression methods.
5. It is easy to implement and is quick to execute.
6. Run-length encoding is supported by most bitmap file formats such as TIFF, BMP and PCX.

Run-Length Coding

WWWWWWWWWWWWBWWWWWWWWWWWWBB
BWWWWWWWWWWWWWWWWWWWWWWWWBW
WWWWWWWWWWWWW

RLE coding:

12W1B12W3B24W1B14W

4. Symbol-Based Coding

In symbol- or token-based coding, an image is represented as a collection of frequently occurring sub-images, called symbols.

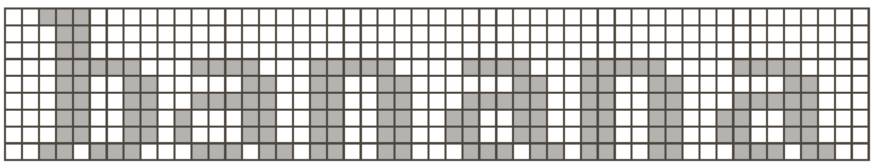
Each symbol is stored in a symbol dictionary

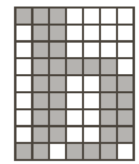
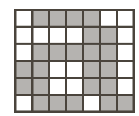
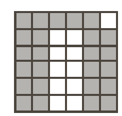
Image is coded as a set of triplets

$\{(x1,y1,t1), (x2, y2, t2), \ldots\}$

35

Symbol-Based Coding



Token	Symbol
0	
1	
2	

Triplet
(0, 2, 0)
(3, 10, 1)
(3, 18, 2)
(3, 26, 1)
(3, 34, 2)
(3, 42, 1)

a b c

FIGURE 8.17 (a) A bi-level document, (b) symbol dictionary, and (c) the triplets used to locate the symbols in the document.

5. Bit-Plane Coding

An m-bit gray scale image can be converted into m binary images by bit-plane slicing. These individual images are then encoded using run-length coding.

Code the bitplanes separately, using RLE (flatten each plane row-wise into a 1D array), Golomb coding, or any other lossless compression technique.

- Let I be an image where every pixel value is n-bit long
- Express every pixel in binary using n bits
- Form n binary matrices (called bitplanes), where the i-th matrix consists of the i-th bits of the pixels of I.

Bit-Plane Coding

Example: Let I be the following 2x2 image where the pixels are 3 bits long

101	110
111	011

The corresponding 3 bitplanes are:

1	1		0	1		1	0
1	0		1	1		1	1

However, a small difference in the gray level of adjacent pixels can cause a disruption of the run of zeroes or ones.

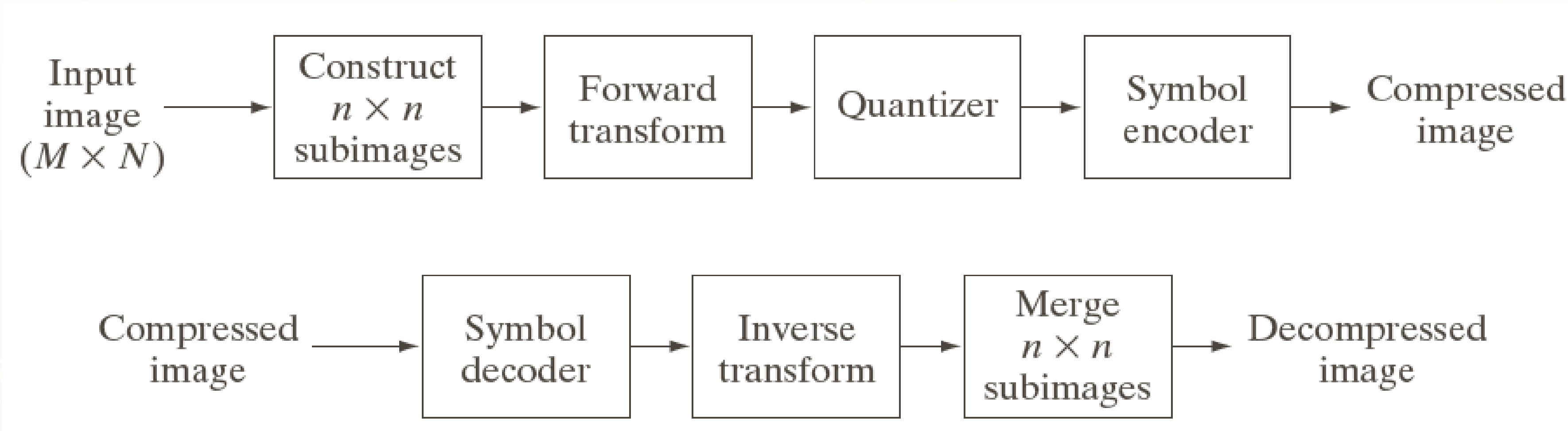
e.g.: Let us say one pixel has a gray level of 127 and the next pixel has a gray level of 128.

In binary: 127 = 01111111

& 128 = 10000000

Therefore a small change in gray level has decreased the run-lengths in all the bit-planes!

Block Transform Coding



a
b

FIGURE 8.21 A block transform coding system: (a) encoder; (b) decoder.

Block Transform Coding

Consider a subimage of size $n \times n$ whose forward, discrete transform $T(u,v)$ can be expressed in terms of the relation

$$T(u,v) = \sum_{x=0}^{n-1} \sum_{y=0}^{n-1} g(x,y) r(x,y,u,v)$$

for $u,v = 0,1,2,...,n\text{-}1.$

Block Transform Coding

Given $T(u,v)$, $g(x,y)$ similarly can be obtained using the generalized inverse discrete transform

$$g(x,y)=\sum_{u=0}^{n-1}\sum_{v=0}^{n-1}T(u,v)s(x,y,u,v)$$

for $x,y=0,1,2,\ldots,n\text{-}1.$

Image Transform

- Two main types:

  - **Orthogonal transform**:
    e.g. Walsh-Hadamard transform, DCT

  - **Subband transform**:
    e.g. Wavelet transform

Orthogonal Transform

- Orthogonal matrix **W**

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} w_{11} & w_{12} & w_{13} & w_{14} \\ w_{21} & w_{22} & w_{23} & w_{24} \\ w_{31} & w_{32} & w_{33} & w_{34} \\ w_{41} & w_{42} & w_{43} & w_{44} \end{pmatrix} \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{pmatrix}$$

➔ C=W. D

- Reducing redundancy
- Isolating frequencies

Walsh-Hadamard Transform (WHT)

$$H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$$

$$H_1 = [1] \qquad H_2 = \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

Discrete Cosine Transform (DCT)

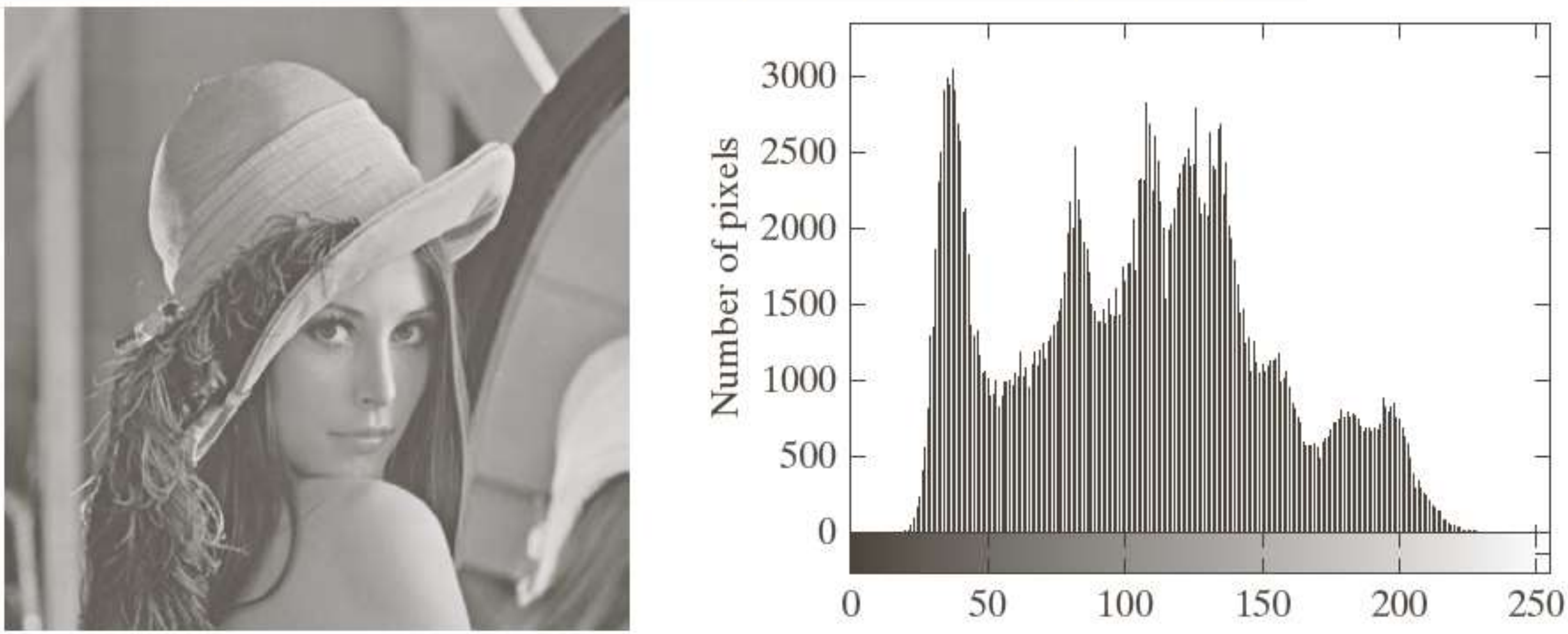
$$r(x,y,u,v) = s(x,y,u,v)$$
$$= \alpha(u)\alpha(v)\cos\left[\frac{(2x+1)u\pi}{2n}\right]\cos\left[\frac{(2y+1)v\pi}{2n}\right]$$

$$\text{where } \alpha(u/v) = \begin{cases} \sqrt{\frac{1}{n}} & \text{for } u/v = 0 \\ \sqrt{\frac{2}{n}} & \text{for } u/v = 1, 2, \ldots, n\text{-}1 \end{cases}$$

Example

a b

FIGURE 8.9 (a) A 512 × 512 8-bit image, and (b) its histogram.



In each case, 50% of the resulting coefficients were truncated and taking the inverse transform of the truncated coefficients arrays.

RMSE = 2.32 RMSE = 1.78 RMSE = 1.13

a	b	c
d	e	f

FIGURE 8.24 Approximations of Fig. 8.9(a) using the (a) Fourier, (b) Walsh-Hadamard, and (c) cosine transforms, together with the corresponding scaled error images in (d)–(f).

Subimage Size Selection

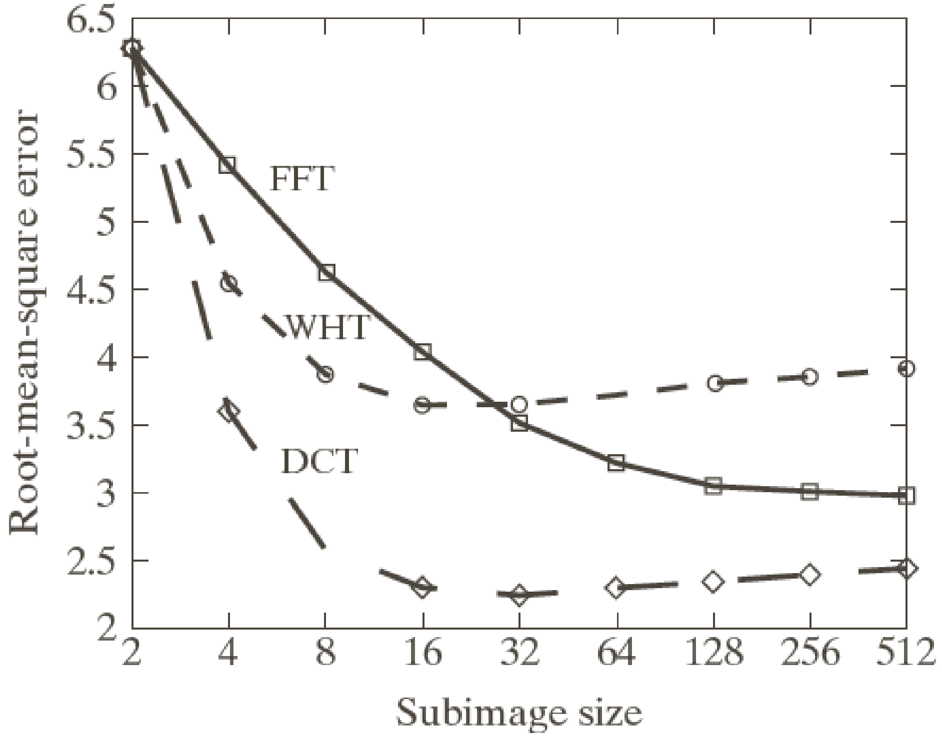


FIGURE 8.26 Reconstruction error versus subimage size.

Fact about JPEG Compression

- JPEG stands for Joint Photographic Experts Group
- Used on 24-bit color files.
- Works well on photographic images.
- Although it is a lossy compression technique, it yields an excellent quality image with high compression rates.

Fact about JPEG Compression

- It defines three different coding systems:

  1. a lossy baseline coding system, adequate for most compression applications

  2. an extended coding system for greater compression, higher precision, or progressive reconstruction applications

  3. A lossless independent coding system for reversible compression

Steps in JPEG Compression

1. (Optionally) If the color is represented in RGB mode, translate it to YUV.

2. Divide the file into 8 X 8 blocks.

3. Transform the pixel information from the spatial domain to the frequency domain with the Discrete Cosine Transform.

4. Quantize the resulting values by dividing each coefficient by an integer value and rounding off to the nearest integer.

5. Look at the resulting coefficients in a zigzag order. Do a run-length encoding of the coefficients ordered in this manner. Follow by Huffman coding.

Discrete Cosine Transform

- The DCT transforms the data from the spatial domain to the frequency domain.
- The spatial domain shows the amplitude of the color as you move through space
- The frequency domain shows how quickly the amplitude of the color is changing from one pixel to the next in an image file.

Step 3: DCT

- The frequency domain is a better representation for the data because it makes it possible for you to separate out – and throw away – information that isn't very important to human perception.

- The human eye is not very sensitive to high frequency changes – especially in photographic images, so the high frequency data can, to some extent, be discarded.

Step 3: DCT

- The color amplitude information can be thought of as a wave (in two dimensions).
- You're decomposing the wave into its component frequencies.
- For the 8 X 8 matrix of color data, you're getting an 8 X 8 matrix of coefficients for the frequency components.

Step 4: Quantize the Coefficients Computed by the DCT

- The DCT is lossless in that the reverse DCT will give you back exactly your initial information (ignoring the rounding error that results from using floating point numbers.)

- The values from the DCT are initially floating-point.

- They are changed to integers by quantization.

Step 4: Quantization

- Quantization involves dividing each coefficient by an integer between 1 and 255 and rounding off.

- The quantization table is chosen to reduce the precision of each coefficient to no more than necessary.

- The quantization table is carried along with the compressed file.

Step 5: Arrange in "zigzag" order

- This is done so that the coefficients are in order of increasing frequency.
- The higher frequency coefficients are more likely to be 0 after quantization.
- This improves the compression of run-length encoding.
- Do run-length encoding and Huffman coding.

FIGURE 8.32 Two JPEG approximations of Fig. 8.9(a). Each row contains a result after compression and reconstruction, the scaled difference between the result and the original image, and a zoomed portion of the reconstructed image.

Compression Performance Measure

$$\text{Compression Ratio (CR)} = \frac{\text{No. of samples x Bit depth per sample}}{\text{Compressed file size}}$$

Percent Root-mean-square Difference

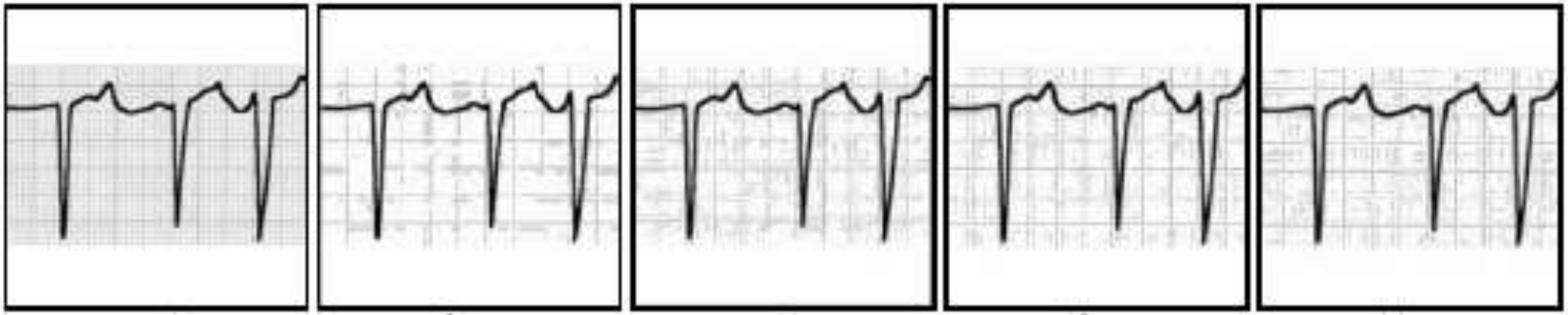
$$PRD = \sqrt{\frac{\sum_{i=1}^{n}\left(x_o(i) - x_r(i)\right)^2}{\sum_{i=1}^{n} x_o^2(i)}} \times 100$$

Where n is the total number of samples, $x_o(i)$ is the original signal and $x_r(i)$ is its reconstruction.

Example: ECG Image Compression

- ECG signal was treated as image recorded on ECG paper where ‘Big blocks’ (5mm on their side) are useful to detect heart rate.
- Performance was evaluated using peak signal to noise ratio (PSNR).
- Coiflet-2 wavelet was found as the best performer in statistical measures.

ECG Image Compression Results (CR=20:1)



Original Haar Daubechies-5 Coiflet-2 Biorthogonal-4.4

Example: Fingerprint Compression

- The US Federal Bureau of Investigation (FBI) has collected about 30 million sets of fingerprints between 1924 and today.

- Fingerprint images digitized at a resolution of 500 pixels per inch for a pair of hands then require about 6 Mbytes of storage.

- Digitizing the FBI's current archive would result in more than 200 Terabytes of data.

- The solution is to compress this data before its storage, transmission and decompress it at the receiver for extensive use.

- Our objective is to discuss the important features of existing methods in compression of fingerprints and compare their performance.

Compression Technique

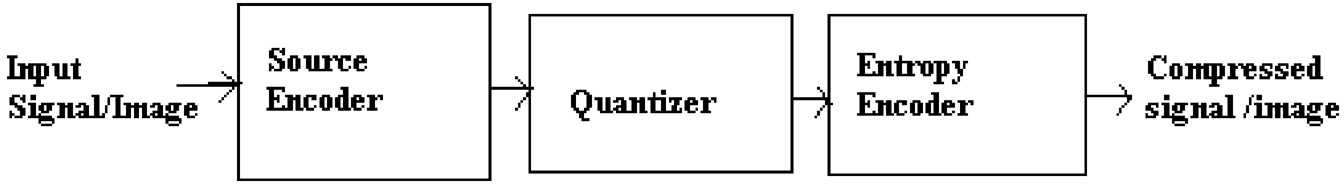
- The overall goal of compression is to represent an image with the smallest possible number of bits, thereby speeding transmission and minimizing storage requirements.

- Compression is of two types-lossless and lossy

- In lossless compression schemes the reconstructed image, after compression, is numerically identical to the original image but it can only achieve modest amount of compression.

- Lossy schemes, though it does not permit perfect reconstruction of the original image, can provide satisfactory quality at a fraction of the original bit rate.

Lossy Image Compression System



- Source Encoder: Decomposes the image into several images for separate processing, typically by linear transformation to concentrate energy in a few coefficients.

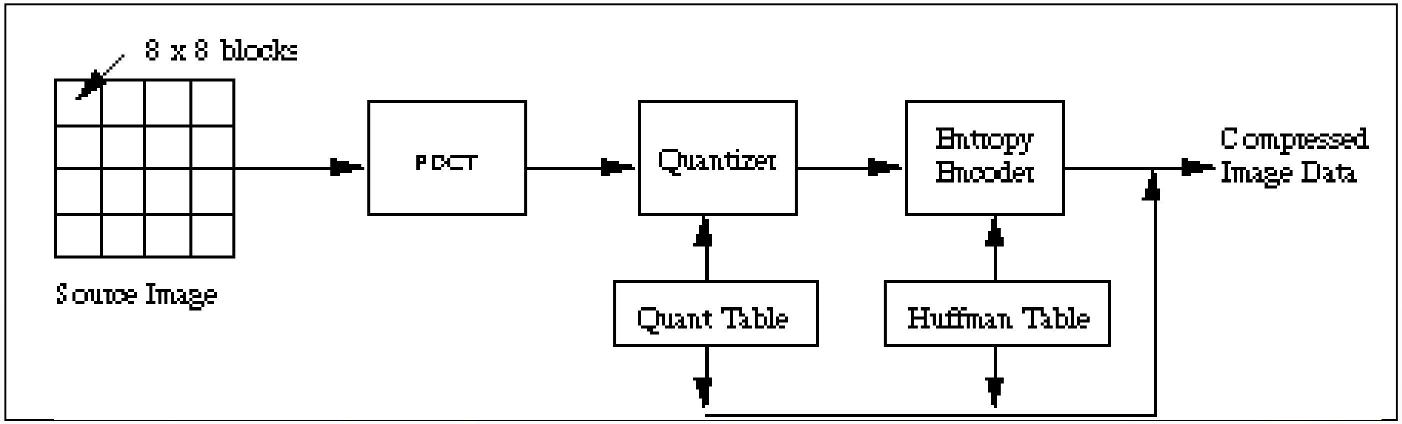
- Quantizer: Reduces the number of bits needed to store the transformed coefficients by reducing the precision of those values making this system lossy.

- Entropy Encoder: Compresses the quantized values using appropriate code based on probability for each quantized value.

Compression Techniques Applied

- Joint Photographic Experts Group (JPEG) standard
- Embedded Zerotree Wavelet (EZW) scheme
- JPEG2000

JPEG Standard

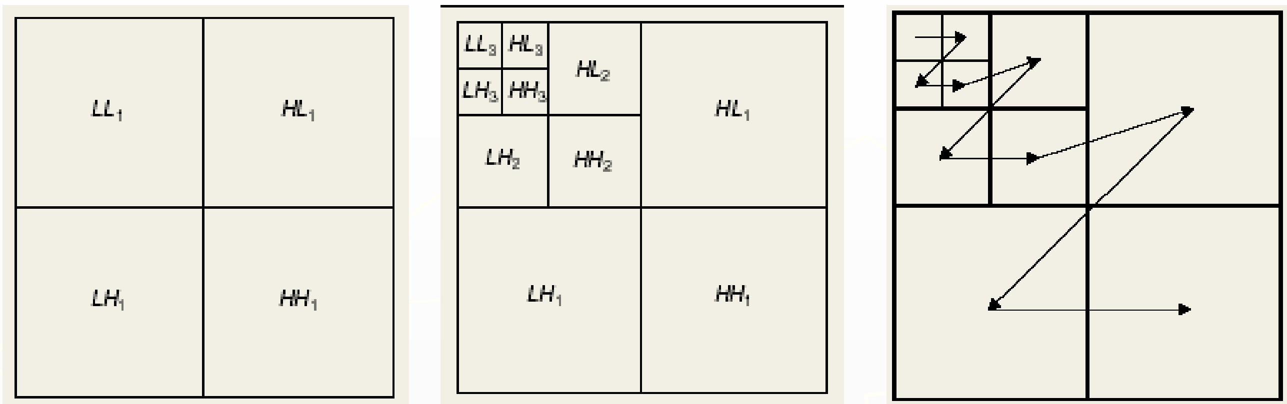


- Encoders and decoders are Discrete Cosine Transform (DCT) based.

- Baseline JPEG coder divides image into 8x8 blocks and transform each block with DCT.

- The transformed blocks are quantized with a uniform scalar quantizer, zig-zag scanned and entropy coded with Huffman coding.

EZW Scheme



- Wavelet transform can decompose a signal into various subbands using octave-band decomposition with wavelet coefficients along different frequency bands.
- The zerotree encoding is based on the hypothesis that if a wavelet coefficient at a coarse scale is insignificant with respect to a given threshold T, then all wavelet coefficients of the same orientation in the same spatial location at finer scales are likely to be insignificant with respect to T.
- This algorithm encodes the tree structure obtained by discarding many insignificant coefficients at higher frequency subbands.

JPEG2000 Scheme

- Based on the dyadic discrete wavelet transform (DWT)
- First step is color transform.
- Color Planes are divided into equally sized blocks called tiles.
- Wavelet transform is applied to the tiles before encoding.
- Wavelet transform can be followed by a quantization step.

Results

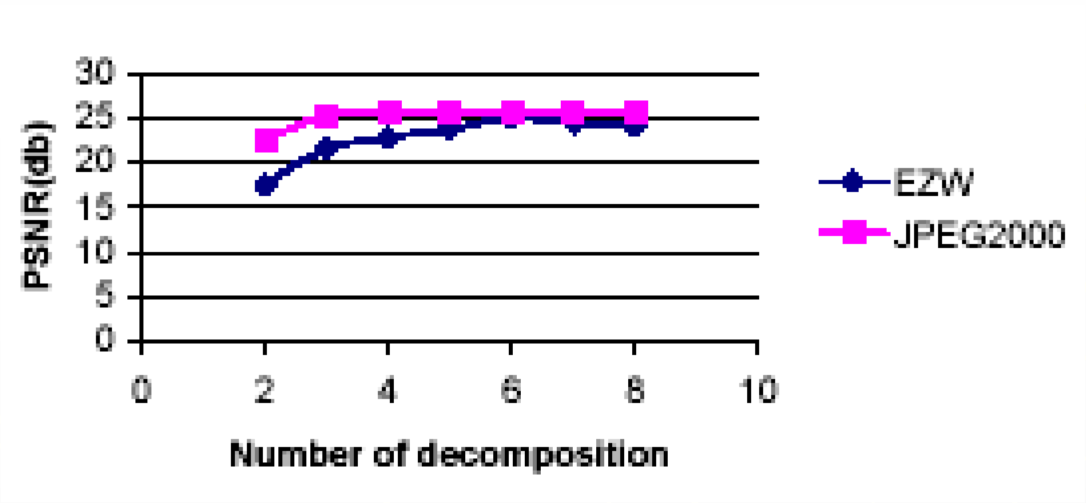
- Our test image was scanned at 500 pixels per inch with 256 levels of grayscale information per pixel with dimension 512x512 pixels.

- Objective quality of the test image was measured by the peak signal to noise ratio (PSNR) which can be defined as

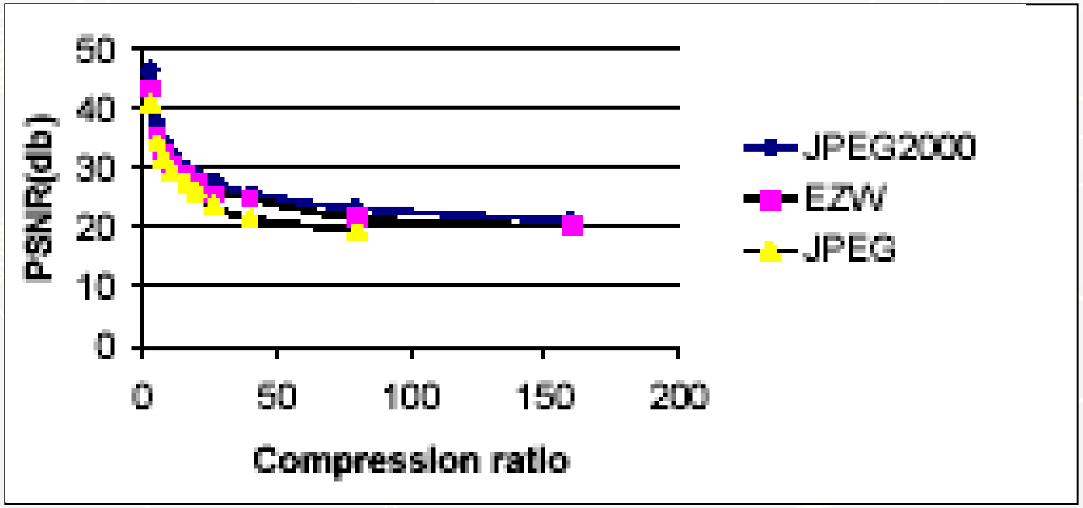
$$PSNR(dB) = 10\log_{10}(255^2/MSE)$$

where MSE stands for mean-square error.

Results



PSNR for different number of decompositions with CR = 40:1



Comparison of PSNR values for 6 decomposition levels

Results : Compressed image with CR=40:1

Original image

JPEG

EZW

JPEG2000

Conclusion

- DCT based JPEG has higher processing speed than the other two schemes discussed.

- At higher compression ratio DCT shows blocking artifacts.

- Blocking artifacts are absent in reconstructed images by wavelet based EZW and JPEG2000 methods.

Questions???

Thank You!!!